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Hypertext Transfer Protocol (HTTP/1.1): Message Syntax and Routing

Abstract

The Hypertext Transfer Protocol (HTTP) is a stateless application-level protocol for distributed, collaborative, hypertext information systems. This document provides an overview of HTTP architecture and its associated terminology, defines the "http" and "https" Uniform Resource Identifier (URI) schemes, defines the HTTP/1.1 message syntax and parsing requirements, and describes related security concerns for implementations.

Status of This Memo

This is an Internet Standards Track document.

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1. Introduction

The Hypertext Transfer Protocol (HTTP) is a stateless application-level request/response protocol that uses extensible semantics and self-descriptive message payloads for flexible interaction with network-based hypertext information systems. This document is the first in a series of documents that collectively form the HTTP/1.1 specification:

1. "Message Syntax and Routing" (this document)
2. "Semantics and Content" [[RFC7231](#)]
3. "Conditional Requests" [[RFC7232](#)]
4. "Range Requests" [[RFC7233](#)]
5. "Caching" [[RFC7234](#)]
6. "Authentication" [[RFC7235](#)]

This HTTP/1.1 specification obsoletes [RFC 2616](#) and [RFC 2145](#) (on HTTP versioning). This specification also updates the use of CONNECT to establish a tunnel, previously defined in [RFC 2817](#), and defines the "https" URI scheme that was described informally in [RFC 2818](#).

HTTP is a generic interface protocol for information systems. It is designed to hide the details of how a service is implemented by presenting a uniform interface to clients that is independent of the types of resources provided. Likewise, servers do not need to be aware of each client's purpose: an HTTP request can be considered in isolation rather than being associated with a specific type of client or a predetermined sequence of application steps. The result is a protocol that can be used effectively in many different contexts and for which implementations can evolve independently over time.

HTTP is also designed for use as an intermediation protocol for translating communication to and from non-HTTP information systems. HTTP proxies and gateways can provide access to alternative information services by translating their diverse protocols into a hypertext format that can be viewed and manipulated by clients in the same way as HTTP services.

One consequence of this flexibility is that the protocol cannot be defined in terms of what occurs behind the interface. Instead, we are limited to defining the syntax of communication, the intent of received communication, and the expected behavior of recipients. If the communication is considered in isolation, then successful actions